

Differentiation 5

Review

1. B as a vector can be expressed as $\begin{pmatrix} 7 \sin 30 \\ 7 \cos 30 \end{pmatrix}$.

C as a vector can be expressed as $\begin{pmatrix} -5 \cos 10 \\ 5 \sin 10 \end{pmatrix}$

Then,

$$\begin{aligned} |AC| &= |\mathbf{B} + \mathbf{C}| \\ &= \left| \begin{pmatrix} 7 \sin 30 - 5 \cos 10 \\ 7 \cos 30 + 5 \sin 10 \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} -1.4240387651 \\ 6.9304187148 \end{pmatrix} \right| \\ &= 7.08 \text{ km} \end{aligned}$$

2. By inspection,

$$3x^4 - 7x^3 + 4x^2 - 8 = (x - 2)(3x^3 - x^2 + 2x + 4)$$

So $3x^4 - 7x^3 + 4x^2 - 8 \div (x - 2) = 3x^3 - x^2 + 2x + 4$

1. $\sin \theta(2 \cos \theta - 1) = 0$

So $\sin \theta = 0 \rightarrow \theta = 0^\circ, 180^\circ$, or $\cos \theta = \frac{1}{2} \rightarrow \theta = 60^\circ, 300^\circ$

2. i. D
ii. E
iii. H
iv. Graph G, gradient function for graph F

$$\begin{aligned} 3. \quad & \int_{-1}^1 (x + 2)^2 dx \\ &= \left[\frac{x^3}{3} + 2x^2 + 4x \right]_{-1}^1 \\ &= \frac{26}{3} \end{aligned}$$

Consolidation

1. i. $5(5x - 2)^4 \times 5 = 25(5x - 2)^4$

ii. $-5(2 + 5x)$

2. i. $4x(x^2 + 1)^3$

ii. $-18x(3x^2 + 5)^{-4}$

3. i. $\frac{5}{2}(5x - 2)^{-\frac{1}{2}}$

ii. $\frac{5}{3}(2 - 5x)^{\frac{4}{3}}$

4. i. $8x(3x^2 + 1)^{\frac{1}{3}}$

ii. $-\frac{12}{5}x(3 - 2x^2)^{-\frac{2}{5}}$

5. i. $3(6x - 5)^{-\frac{1}{2}}$

6. ii. $(-10x + 20)(x^2 - 4x - 3)^{-2}$

7. i. $(-\frac{3}{2}x^2 - \frac{3}{2})(x^3 + 3x)^{-\frac{3}{2}}$

8. i. $y' = 4e^{4x}$

ii. $y' = -2e^{-2x}$

9. i. $\frac{1}{x}$

ii. $\frac{1}{x}$

10. i. $4 \cos 4x$

ii. $-3 \sin 3x$

iii. $\sec^2 x$

11. $f'(x) = -2(2x - 1)^{-2}$

$$f'(1) = -2$$

12. $\frac{dy}{dx} = -3 \cos 3x$

$$\frac{d^2y}{dx^2} = -9 \sin 3x$$

$$= -9y$$

13. $y' = 5x(5x^2 + 16)^{-\frac{1}{2}}$

At $x = 2$, $y' = \frac{5}{3}$.

So the normal to the curve will have gradient of $-\frac{3}{5}$.

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -\frac{3}{5}(x - 2)$$

$$5y - 30 = -3x + 6$$

$$3x + 5y - 36 = 0$$

14. When $x = 4$, $y = \frac{10}{\sqrt{16+9}} = \frac{10}{5} = 2$.

$$y = 10(x^2 + 9)^{-\frac{1}{2}}$$

$$y' = (-10x)(x^2 + 9)^{-\frac{3}{2}}$$

When $x = 4$, $y' = \frac{8}{25}$.

So the tangent at $x = 4$ can be found by

$$y - 2 = \frac{8}{25}(x - 4)$$

$$25y - 50 = 8x - 32$$

$$8x - 25y + 17 = 0$$

When $y = 0$, $x = -\frac{17}{8}$.

When $x = 0$, $y = \frac{17}{25}$.

15. a.

$$\frac{dr}{dt} = 0.2$$

b. i. $P = 2\pi r$

$$\begin{aligned}\text{ii. } \frac{dP}{dt} &= \frac{dP}{dr} \times \frac{dr}{dt} \\ &= 2\pi \times 0.2 \\ &= 0.4\pi\end{aligned}$$

c. i. $A = \pi r^2$

$$\begin{aligned}\text{ii. } \frac{dA}{dt} &= \frac{dA}{dr} \times \frac{dr}{dt} \\ &= 2\pi r \times 0.2 = 0.4\pi r\end{aligned}$$

When $r = 10$, area is increasing by $4\pi \text{cms}^{-1}$.

d.

$$\begin{aligned}A' &= 20 \\ &= 0.4\pi r \\ \therefore r &= \frac{50}{\pi} = 15.9\end{aligned}$$

e. i.

$$V = x^3$$

ii. Because the rate of change of the side length is -3 : it is decreasing at a rate of 3, so $\frac{dx}{dt} = -3$.

iii.

$$\begin{aligned}\frac{dV}{dt} &= -9x^2 \\ &= -225 \text{ when } x = 5.\end{aligned}$$

Exam Questions

1. If $f(x)$ is always decreasing, $f'(x)$ will always be negative.

$$\begin{aligned}f(x) &= 6(x^2 + a)^{-1} \\ f'(x) &= -12x(x^2 + a)^{-2}\end{aligned}$$

Since $x > 0$, x is always positive, so $-12x$ will always be negative. Therefore, f is a decreasing function for all values of x because the gradient of the function is always negative.

2.

$$y = (\cos x)^{-1}$$

$$\begin{aligned}y' &= -(\cos x)^{-2} \times -\sin x \\ &= \sin \frac{x}{\cos^2 x} \\ &= \left(\sin \frac{x}{\cos} \right) \times \frac{1}{\cos} \\ &= \sec x \tan x\end{aligned}$$

3. a.

$$\begin{aligned}
 y &= (3 + 4e^{x^2})^{\frac{1}{2}} \\
 u &= 3 + 4e^{x^2} \\
 &\rightarrow y = u^{\frac{1}{2}} \\
 \frac{dy}{du} &= \frac{1}{2}u^{-\frac{1}{2}} \\
 \frac{du}{dx} &= 8xe^{x^2} \\
 \therefore \frac{dy}{dx} &= \frac{1}{2}(3 + 4e^{x^2})^{-\frac{1}{2}} \times 8xe^{x^2} \\
 &= \frac{8xe^{x^2}}{2(3 + 4e^{x^2})^{\frac{1}{2}}} \\
 &= \frac{4xe^{x^2}}{\sqrt{3 + 4e^{x^2}}}
 \end{aligned}$$

b. When $x = \alpha$,

$$\begin{aligned}
 y - \sqrt{3 + 4e^{\alpha^2}} &= \left(\frac{4ae^{a^2}}{\sqrt{3 + 4e^{a^2}}} \right) (x - \alpha) \\
 y &= \left(\frac{4ae^{a^2}}{\sqrt{3 + 4e^{a^2}}} \right) (x - \alpha) + \sqrt{3 + 4e^{\alpha^2}}
 \end{aligned}$$

Then, we can say that at $(0, 0)$:

$$\begin{aligned}
 0 &= -\frac{4\alpha^2 e^{\alpha^2}}{\sqrt{3 + 4e^{\alpha^2}}} + \sqrt{3 + 4e^{\alpha^2}} \\
 &= 3 + 4e^{\alpha^2} - 4\alpha^2 e^{\alpha^2} \\
 &= 3 + 4e^{\alpha^2}(1 - \alpha^2)
 \end{aligned}$$

By deep thinking:

$$4\alpha^2 e^{\alpha^2} - 4e^{\alpha^2} - 3 = 0$$

and so

$$4x^2 e^{x^2} - 4e^{x^2} - 3 = 0$$

when $x = a$.

4.

$$\begin{aligned}
 y' &= -6 \sin 3x - 3 \\
 \sin 3x &= -\frac{1}{2} \\
 3x &= \frac{7\pi}{6}, \frac{19\pi}{6} \\
 x &= \frac{7\pi}{18}, \frac{19\pi}{18}
 \end{aligned}$$

So $\beta = \frac{7\pi}{18}$ and $\lambda = \frac{19\pi}{18}$